

CS 180 • Project 1

[Overview](#) [Final Results](#) [Test Results](#)UC BERKELEY • CS180/280A: INTRO TO COMPUTER VISION AND
COMPUTATIONAL PHOTOGRAPHY

Colorizing the Prokudin-Gorskii Glass Plates

This project reconstructs color images from stacked grayscale negatives by splitting the three channels, aligning them with either a single-scale search or a coarse-to-fine pyramid, and then composing the final RGB result.

[Single-scale alignment](#)[Pyramid alignment](#)[GitHub Pages gallery](#)**TASK**

Split and align stacked channels

The input image contains the blue, green, and red exposures stacked vertically. The notebook first splits the scan into three equal parts, trims the borders, and then searches for the best translation that aligns the green and red channels to blue.

STRATEGY

Use single-scale for small scans, pyramid for large scans

The three special images, **cathedral**, **monastery**, and **tobolsk**, are shown as single-scale results. The remaining images use the pyramid method because the displacement is larger and the full-size scans are more expensive to search directly.

PRESENTATION

Compare methods side by side

The test set is displayed row by row so you can compare the signal aligner result and the pyramid result for the same image immediately. The layout mirrors the project-webpage style expected for CS180: concise explanation, grouped sections, and image-first presentation.

PART 1**Final Results**[14 images](#)

The three low-resolution examples at the top were aligned using a single-scale signal aligner. The rest of the gallery uses the pyramid pipeline, which is better suited to the large TIFF scans and larger displacements. Border pixels are ignored during scoring so that the wrap-around artifacts from shifting do not affect the match.

Signal aligner

Signal aligner

By L2



Red offset: dx=3, dy=12; Green offset: dx=2, dy=5

By NCC



Red offset: dx=3, dy=12; Green offset: dx=2, dy=5

cathedral.jpg

Signal aligner

By L2



Red offset: dx=2, dy=3; Green offset: dx=2, dy=-3

By NCC



Red offset: dx=2, dy=3; Green offset: dx=2, dy=-3

monastery.jpg

Signal aligner

By L2



Red offset: dx=3, dy=6; Green offset: dx=3, dy=3

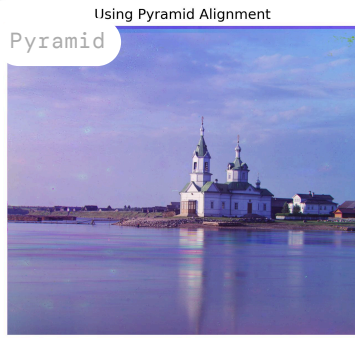
By NCC



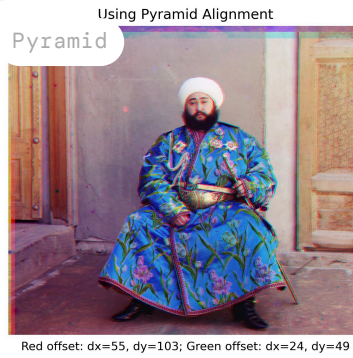
Red offset: dx=3, dy=6; Green offset: dx=3, dy=3

tobolsk.jpg

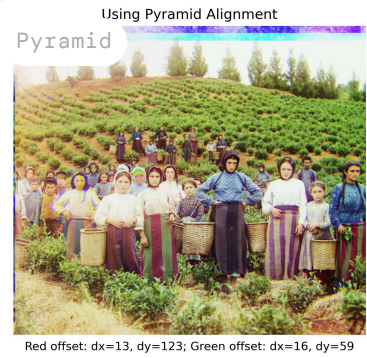
Pyramid



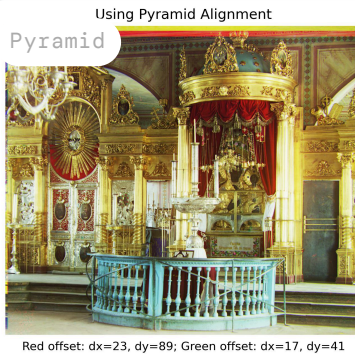
church.jpg



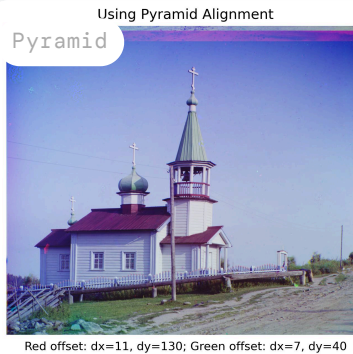
emir.jpg



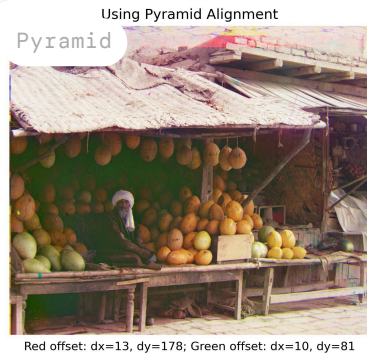
harvesters.jpg



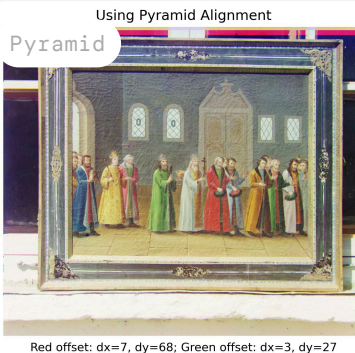
icon.jpg



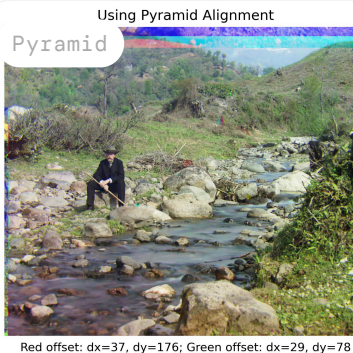
ilemselga.jpg



melons.jpg



religious_painting.jpg



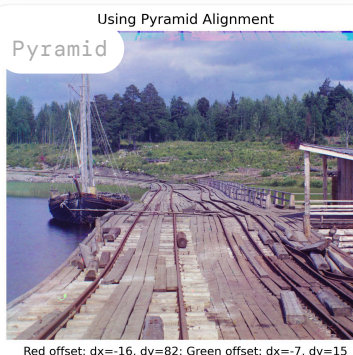
self_portrait.jpg



siren.jpg



three_generations.jpg



wharf.jpg

For the test images, each row shows the same input processed by two methods. This makes it easy to compare whether the signal aligner is already sufficient or whether the pyramid method produces a cleaner final alignment.

TEST2

Signal aligner



Red offset: dx=-4, dy=15; Green offset: dx=2, dy=2

test2_jpg.jpg

By NCC



Red offset: dx=-11, dy=15; Green offset: dx=2, dy=2

Pyramid



Red offset: dx=-165, dy=293; Green offset: dx=31, dy=38

test2_tif.jpg

Using Pyramid Alignment

TEST3

Signal aligner



Red offset: dx=-2, dy=7; Green offset: dx=-1, dy=2

test3_jpg.jpg

By NCC



Red offset: dx=-2, dy=7; Green offset: dx=-1, dy=2

Pyramid



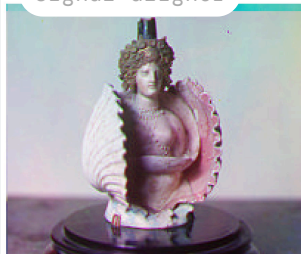
Red offset: dx=-25, dy=121; Green offset: dx=-14, dy=34

test3_tif.jpg

Using Pyramid Alignment

TEST4

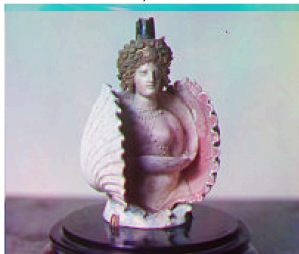
Signal aligner



Red offset: dx=1, dy=5; Green offset: dx=1, dy=0

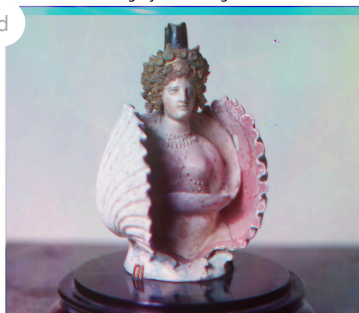
test4_jpg.jpg

By NCC



Red offset: dx=1, dy=5; Green offset: dx=1, dy=0

Pyramid



Red offset: dx=12, dy=77; Green offset: dx=8, dy=10

test4_tif.jpg

Using Pyramid Alignment

Hosted from the **web** branch on GitHub Pages.

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